

Hardware Product Planning Essentials For Start-ups And SMEs

Most start-ups and SMEs assume the prototype can be quickly converted to a regular product, with little effort, which is a fallacy. Converting a prototype or an early-stage version to a manufacturable product takes lots of effort, time, and money



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Planning is critical for hardware product start-ups and SMEs. By planning I mean the technical aspects of the product and the execution. In my experience, in case of most start-ups and SMEs, I have seen, planning has been rather transactional, in the sense that they keep doing it as they go along—without a final end date as a goal. When a new product is planned there are three aspects which have to be nailed and should not be compromised. These are:

1. Costing of bill of material (BoM) for various volumes and supply chains for Class A (most critical and high cost) parts.

2. Manufacturing partner who will manufacture and the type of manufacturing needed. Typically, there are three types of manufacturing, start-ups and SMEs can follow:

- (a) Get the PCBs soldered (no further processing) and components supplied by the product company. Normally this type is known as 'Job Work.'

- (b) From the supplied BoM assemble the PC boards, test, and deliver. This is normally known as 'PCB Assembly.' This step may involve testing of PCBs if the EMS vendor has 'flying probe' testers.

- (c) Complete product manufacturing with start-ups providing only the documentation. This type is known as 'Box Build.'

3. International and national standards that the product/solutions should meet.

Most of the times, these three items are given low priority and addressed only in the last minute, which leads to utter chaos. I am not a votary of huge project planning, but if the essentials are not covered things are bound to fail. The above three issues, I have seen, come to haunt as the project progresses.

Why planning is critical?

When a customer buys a hardware based product, factors that are mainly looked at are cost, performance, and support. I have